TOL: what it can do today

Version 0.4.6 for macOS and Windows. A transparent, neutral window that shows which companies a computer is sending data to. It only watches. It does not block, change or read anything.

What it does

Capability	macOS	Windows
Live list of destinations: who the computer is connected to, which program opened each connection, and when	**Yes** Run on a real Intel Mac.	**Yes** Passes automated tests on Windows machines. **Not yet run on a real PC.**
Amount of data sent and received per destination	**Yes**	**No** Windows does not give ordinary programs per-connection counts. TOL shows n/a and says why on screen.
Names the owner of an address (public registry lookup, only when the user asks; one IP address is sent, nothing else)	**Yes**	Built and tested by machine. **Not yet run on a real PC.**
Receipts: each lookup saved verbatim with its hash, in a chain that breaks if any entry is edited or removed	**Yes**	Built and tested by machine.
Recording, and an evidence package: the recording, the receipts, device details, a manifest of every file's fingerprint, and a checking script	**Yes** Made and checked end to end on a real Mac.	Built into the packaged app; the build is tested on Windows machines.
Checking script (verify.py): anyone can run it with plain Python and no TOL. It reports whether any file was changed or added and whether the chains are intact	**Yes** Passes on a package made today. Rejected a file added afterwards.	Same script; tested by machine.
Trusted timestamp (optional): only a fingerprint is sent to an independent authority (DigiCert). TOL's own checker verifies the authority's signature and certificate chain, with nothing extra to install	**Yes** Verified on a real Mac against DigiCert.	Same checker. It reads the Windows certificate store, a route not yet run on a real PC.
Small separate window to keep beside other work (Pop out)	**Yes** Reopens reliably on a real Mac.	Finds Edge or Chrome. Not yet run on a real PC.

What it does not do

- **It does not read content.** It sees the destination, the program and the amount, not the words, files or pages inside.
- **It does not block or change anything.** Turning it off leaves the computer exactly as it was.
- **It can miss very short connections** that open and close between two checks, a few seconds apart.
- **A named owner is not always the vendor.** A registry names who holds an address block, which may be a cloud or hosting company.
- **It is not a tamper-proof device.** The chains show records were not edited after they were written, not that the computer was untouched. Without a trusted timestamp, times come from the computer's own clock.
- **Timestamp limits.** The checker does not test whether the authority's certificate was revoked. It handles RSA signatures with SHA-2 (DigiCert and Sectigo); other signature types are reported as not verified. It is a standard timestamp, not a legally "qualified" one.

How it was checked

Evidence	What it covers
Real Intel Mac, TOL 0.4.6	Live view, recording, evidence package, trusted timestamp and the checking script all run on a real computer. The check passed on a package made that day.
Real timestamp answers	Saved answers from DigiCert and Sectigo verify, including their full certificate chains. A third authority (FreeTSA) uses a different signature type and is correctly reported as not verified.
Attack tests	Every byte of what the authority signed, the signature and the signing certificate was changed one at a time on two real answers. None of the changed answers was accepted.
Over 170 automated tests	Run on every change, on a Mac and on Windows test machines, before a release is built. They include altered and forged packages.
Written method	A plain-language description of what is recorded, how, and what it does not prove (docs/METHOD.md).

Not yet done

- **Real-PC test of the Windows version.** Everything marked above as not yet run on a real PC.
- **Real-computer test of the Apple-chip Mac build.** It is built and passes automated tests; no person has run it.
- **Code signing.** Both apps are unsigned, so a Mac or Windows computer shows a first-time warning, and some managed computers may not allow them.
- **Independent review.** No outside party has audited TOL or its method.
- **Certificate revocation checking** and support for more timestamp signature types.

Using it

TOL runs on the person's own computer and shows its dashboard only on that computer (address 127.0.0.1). It has no accounts and no analytics, and it does not phone home. Nothing leaves the computer except two optional actions, each started by the user: a registry lookup (one public IP address) and a timestamp request (one 32-byte fingerprint). Data is kept in a folder on the user's own computer.

Overview page: rickwatson5166.github.io/tol-info